BSc. (Hons.) Engineering in Information and Communication Engineering (ICE)

Literature survey

EEE3305-Engineering Research (Embedded Systems) Project (3C)

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Alarm Clock Development using PIC Microcontroller and RTC

Review of Existing Alarm Clocks

1. Commercial Alarm Clocks:
Analog Clocks: Classic, reliable, and often aesthetically pleasing, analog clocks lack many features like snooze, display backlight, and radio integration. Their analog display can be harder to read at night or from afar.
Digital Clocks: Offer diverse features like radio, multiple alarms, timer, and even temperature sensors. However, their reliance on mains power makes them vulnerable to power outages, and their displays can be harsh on the eyes at night.
Smartphones: With built-in clocks, alarms, and endless app possibilities, smartphones have become ubiquitous alarm-keeping devices. Their reliance on battery and potential for notifications interrupting sleep are drawbacks.

2. Microcontroller-Based Alarm Clocks:

DIY Enthusiasts: Several online projects showcase various designs using PIC microcontrollers and different RTC modules. These offer customization and learning opportunities, but require technical expertise and breadboard setups.
Commercially Available Kits: Ready-made kits provide user-friendly introductions to building microcontroller-based alarm clocks. However, they often have limited features and lack the flexibility of designing from scratch.

3. Key Technologies in Existing Solutions:
Real-Time Clocks (RTCs): Essential for maintaining accurate time even during power outages.
Displays: LCD displays are popular for displaying time and alarm settings, with options like 16x2 and 2x16 offering varying information visibility.
Buzzers and LEDs: Audible and visual cues for alarm activation, with piezo buzzers and LEDs being common choices.
Input Devices: Buttons or keypads allow users to set time, alarm parameters, and adjust settings.

4. Limitations and Areas for Improvement:
Technical Complexity: Existing microcontroller-based designs can be daunting for beginners with limited electronics knowledge.
Feature Sets: Existing kits offer basic functionalities, while DIY projects might lack sophisticated features like radio integration or smartphone connectivity.
Energy Efficiency: Microcontroller-based clocks usually require external power supplies, increasing their energy consumption.
User Interface: Some designs have cumbersome button menus or lack intuitive time/alarm setting interfaces.

5. Your Project's Potential Contribution:
Utilizing a PIC microcontroller and RTC, has the potential to address some of these limitations and contribute to the field of alarm clock technology.
By focusing on:
Ease of Use: Simplifying the design and user interface for broader accessibility.
Advanced Features: Incorporating functionalities like smartphone connectivity, customizable sound options, or light-sensitive brightness adjustments.
Energy Efficiency: Utilizing low-power components and sleep modes for extended battery life.

Previous Solutions using PIC Microcontrollers and RTCs

PIC microcontrollers, known for their affordability, ease of use, and wide range of features, have become popular choices for DIY alarm clock projects. Several existing projects demonstrate the feasibility of using PICs and RTCs for alarm clock development:

Simple Alarm Clock: This basic design utilizes a PIC16F84A microcontroller and an DS1307 RTC module to display time and trigger an alarm at a set time. The project demonstrates core functionalities like timekeeping, alarm setting, and buzzer output.

Simple Alarm Clock using PIC16F84A microcontroller and DS1307 RTC module
Advanced Alarm Clock with Snooze and Multiple Alarms: This project expands on the basic design by adding features like a snooze button and multiple alarms.

Radio-Controlled Alarm Clock: This project incorporates a radio receiver module to synchronize the clock with an external time signal, ensuring high accuracy.
These examples showcase the versatility of PIC microcontrollers and RTCs in alarm clock design. They provide a starting point for our project, allowing us to build upon existing solutions and incorporate additional features or functionalities.

Comparison of Methodologies and Technologies

Several methodologies and technologies can be employed in alarm clock development using PIC microcontrollers and RTCs. Here's a comparison of some key aspects,

	Simple Alarm Clock	Advanced Alarm Clock	Radio-Controlled Alarm Clock
Microcontroller	PIC16F84A	PIC16F887	PIC18F25K50
RTC Module	DS1307	DS1307	DS1307
Display	LCD	LCD	LCD
User Interaction	Buttons	Buttons	Buttons
Alarm Options	Single Alarm	Multiple Alarms, Snooze	Multiple Alarms, Snooze
Timekeeping Accuracy	Internal Oscillator	Internal Oscillator	Radio Signal Synchronization
Additional Features	None	None	Feature

We can select the appropriate microcontroller, RTC module, and additional components based on our desired functionalities and project constraints.

Challenges and Opportunities

Challenges:

Hardware interfacing: Integrating various components like the microcontroller, RTC, display, buttons, and buzzer requires careful circuit design and programming.
Power management: Optimizing power consumption is crucial for extended battery life in portable alarm clocks.
Software development: Programming the microcontroller to handle timekeeping, user interaction, alarm functionalities, and other features requires knowledge of embedded C programming.

Opportunities:

Customization: PIC microcontrollers offer flexibility for implementing unique features and functionalities, personalizing the alarm clock experience.
Integration with other devices: The project can be expanded to connect with other devices like smartphones or smart home systems for enhanced functionality.
Learning and development: Building an alarm clock provides valuable hands-on experience in electronics, programming, and project management.